

Towards Controlling Whole Body Avatars with Partial Body-Tracking and Environmental Information

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ABSTRACT

In body-tracking-based avatar manipulation, the user's motion is reflected in their avatar, which creates a high level of immersion. Conversely, the user is required to give a detailed performance similar to that of the avatar to be composed.

In this research, we aim to produce avatar motion that reflects the user's intended actions and maintains consistency with the VR environment by inputting the user's posture. The avatar's body motion was generated by inputting the user's motion and VR environmental information into the motion configuration network, which was developed using machine learning.

Keywords: CG avatar, metaverse, interaction.

Index Terms: Computing methodologies—Computer graphics—Graphics systems and interfaces—Virtual reality; Computing methodologies—Computer graphics—Animation—Motion capture; Computing methodologies—Machine learning—Learning paradigms—Supervised learning—Supervised learning by regression

1 INTRODUCTION

A CG avatar is used to represent users in VR, particularly for body-tracking-based avatar manipulation. The purpose of using an avatar is to enhance immersion, because the user's motion is then reflected directly in the VR world.

To achieve a high level of immersion, the avatar must perform the actions that the user intended. Therefore, the user must perform actions similar to those of the avatar. Thus, if the posture information of the whole body cannot be obtained, it is difficult to construct the intended avatar motion through whole-body tracking.

The study by Zielasko et al [1] refers to the case of manipulating an avatar in a sitting position using a joystick and body tilt. In this case, the interpretation of the posture and method of movement varies from user to user. For example, it is difficult to give the user a sense of walking when the avatar is operated in a seated posture, even if it is operated by a controller such as a joystick. Therefore, to give the user the sensation of walking, it is considered preferable to generate whole-body motion from only the partial posture input of the user without using a controller.

The method used by Heck et al.[2] constructed avatar motions from partial posture information. In this method, an avatar's actions were produced by interpolating the lower body motion from one source with the upper body posture from another. However, this method is not intended for avatar actions constructed using body-tracking, and it is difficult to reflect a user's intention in an avatar's

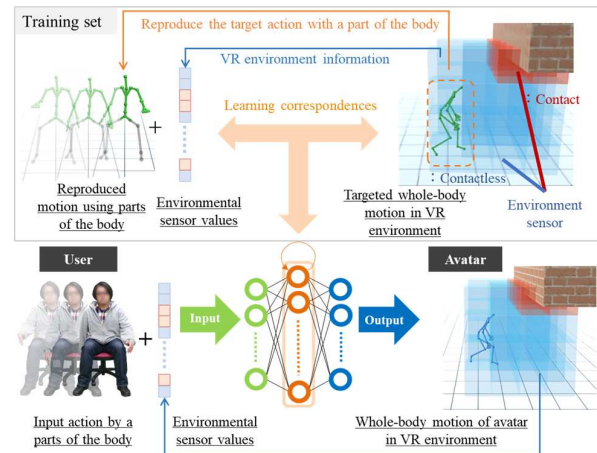


Figure1: Overview of the proposed method

behavior. In addition, a large amount of motion data must be captured to describe clear conditions to configure various motions, which requires considerable effort.

In this paper, we propose a method for using machine learning to construct an avatar's motion based on the user's posture input that reflects the user's intention and is consistent with the VR surroundings. The training set consists of information on the whole body motion of the avatar in the VR environment, the environment around the avatar, and inputs of the user's partial posture that show the intended motion. By learning from the user's posture input and the VR environment in which the avatar is placed, whole-body avatar motion can be produced that is consistent with the VR environment even when there are different operational intentions for a similar posture input depending on the VR environment.

2 METHODS FOR CONFIGURING AVATAR MOTION

2.1 Overview

Figure 1 shows the proposed method. The avatar motion configuration network was constructed using a recurrent neural network. The data of the training set were the motion data of the whole body of the avatar in the target VR environment. The input data were the posture information from any body part of the user intended to actuate the target motion and the environmental sensor information of the avatar. Using the above training set, the three-way correspondence between the temporal features of the body part's motion and the surrounding environmental information, and the whole-body motion was trained using supervised learning.

This paper focuses on the upper trunk as the body part selected for the user to operate, assuming that the user operates the avatar in a sitting posture. To obtain information on the surroundings of the avatar's VR environment, based on [3], a set of cubic areas was placed around the avatar model, and the results of contact estimation between each cubic region and environmental objects were used as environmental sensor information.

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2.2 Creating Training Set

The training set was created via the following procedure.

- (1) Whole-body motions were recorded that were consistent with the target VR environment (or existing motion data were prepared).
- (2) The motion data in step 1 were applied to the avatar model and played back in the target VR environment. While checking the execution image, the user performed actions assuming that they were manipulating the avatar with an arbitrary body part. The user's motion and the avatar's environmental sensor information were recorded at this time.
- (3) The time delay between the visual reproduction of step 2 and the target whole-body motion was corrected, the product of step 1 was used as the training data and the corrected step 2 data were used as the corresponding input data.

2.3 Building the Avatar Motion Configuration Network

This research used recurrent neural networks, a type of neural network that account for the temporal characteristics of time-series data. The recurrent neural networks were constructed using long short-term memory (LSTM) units. The network consisted of three layers: an input layer, an output layer, and an LSTM layer. The total number of input units was 107, which consisted of 51 rotations of 17 upper body joints along three axes and 56 of the avatar's environmental sensor data points. The total number of output units was 156, which consisted of 51 whole body joint rotations along three axes and the avatar's motion information. The LSTM was 1000 layers deep. The backpropagation through time (BPTT) was set to 40 to account for the time dependency between 40 frames. Assuming that the amount of each joint rotation of the intended whole-body motion is $\hat{p}_n$, the amount of each joint rotation output from the network is p_n , and the number of output joints is N . The error function L is expressed by the following equation.

$$L = \frac{1}{N} \sum_{n=1}^N (p_n - \hat{p}_n)^2 \quad (1)$$

To suppress overfitting, we introduced early stopping, which considers learning to be complete when the value obtained from the loss function has increased for a certain number of iterations.

3 EXPERIMENT

We conducted experiments on an avatar walking in a VR environment with an obstacle overhead.

The target motions of the avatar in the VR environment were a half-bent walking motion to avoid overhead obstacles and a normal walking motion. Existing motion capture data were used. In addition, an arm-waving motion intended to manipulate the target motion was performed while in a seated posture, and the upper body motion was recorded using Kinect v2. Simultaneously, we acquired environmental sensor values in the VR environment. The data above were used as the training set. The training set totaled 37,700 frames (approximately 21 min). Using this training data, we constructed the network defined in Section 2.3.

Figure 2 shows the results of configuring the avatar's motion by inputting the upper body motion in the sitting posture and the environmental sensor values into the trained network. The green skeleton in Figure 2 is the joint of the upper body used for input. Additionally, in the figure, the sensors around the avatar detect contact with the object; they are light blue if there is no contact, and red if there is contact. The input motion of the upper body was based on the prerecorded motion data and were not included in the training set. Figure 2 shows that the avatar's walking motion was constructed from the upper body's input of an arm-swinging motion and that the walking motion was constructed from a standing position or a bent-over position depending on the values obtained from the environmental sensor.

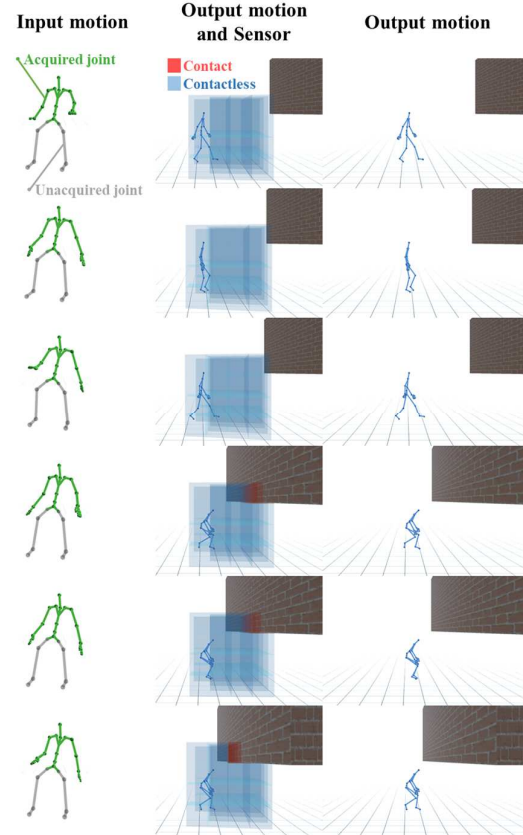


Figure 2: Configuration results of the walking motion.

4 CONCLUSION

This paper proposes a method that uses machine learning to construct avatar behaviors that reflect the user's intentions, based on the user's posture input and information from the VR environment in which the avatar is placed. By conducting experiments using walking motions, we confirmed that the system was able to reflect the user's functional intentions from the motion input of the upper body while in a sitting posture and that it could construct whole-body motion that varied depending on the environment. Future work could involve verifying the results of motion composition for various motions in diverse environments, based on the subjective evaluation of the user regarding whether operational intentions have been accurately reflected.

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REFERENCES

- [1] D.Zielasko, B.E.Riecke, "Can We Give Seated Users in Virtual Reality the Sensation of Standing or Even Walking? Do We Want To?", 2020 IEEE Conference on Virtual Reality and 3D User Interfaces Abstracts and Workshops (VRW), pp. 281-282, 2020.
- [2] R.Heck, L.Kovar, M.Gleicher, "Splicing Upper-Body Actions with Locomotion", Computer Graphics Forum, Vol.25 Issue.3, September 2006 pp.459-466, 2006.
- [3] S.Starke, H.Zhang, T.Komura, J.Saito: "Neural State Machine for Character-Scene Interactions", ACM Transactions on Graphics, 38(6), Article No.178, 2019.